ELEC70043
ELEC60003

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2022

ANALOGUE INTEGRATED CIRCUITS AND SYSTEMS

Tuesday 6 December 10:00 am

Time allowed: 3:00 hours

There are SIX questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : P. Georgiou

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1. (a) (i) Figures 1(a) and 1(b) show two popular biasing schemes typically used in analogue integrated systems. Briefly outline the main feature of each of these circuits.

[4]

(ii) For the bandgap voltage reference circuit of Figure 1(a), show that $\delta V_0 / \delta T = 0$ (where T is temperature) if $(R_2/R_3) ln [I_1/I_2] = 29$ for $V_0 = 1.283\ V$. Assume the temperature coefficient of V_{BE} to be -2.5mV/°C, the collector current of transistor Q_3 is 100µA and the device saturation current is $I_S = 1.2 \times 10^{-13} A$. Boltzmann's constant $k = 1.38 \times 10^{-23} J/K$ and the electron charge is $q = 1.6 \times 10^{-19} C$.

[7]

(b) Figure 1(c) below shows a single stage of a differential amplifier.

(i) Draw a circuit schematic for the current source I_0 that ensures the differential gain is independent of temperature and power supply variation. Include any start up circuits that may be required.

[4]

(ii) For your design, show that the gain bandwidth product is independent of temperature.

[3]

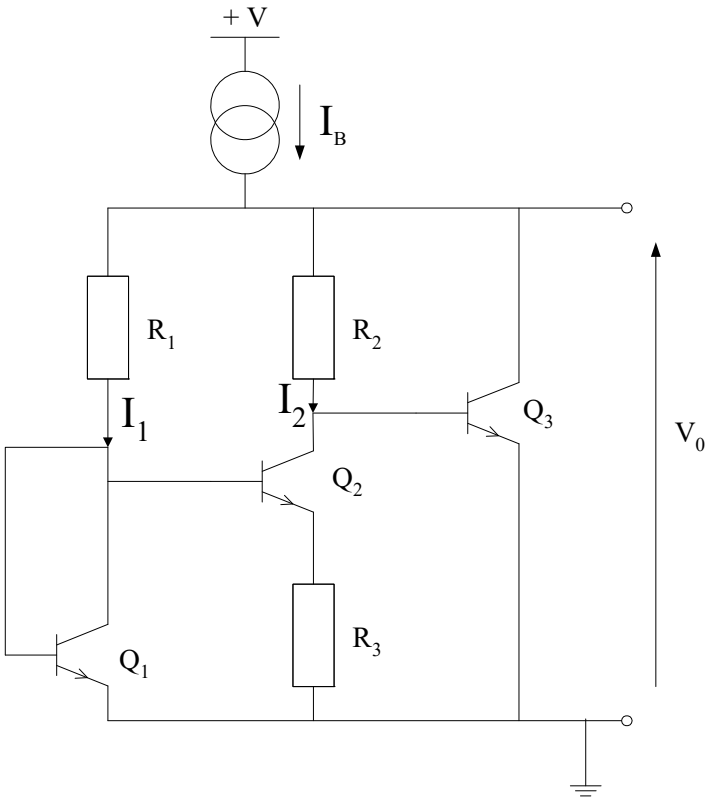


Figure 1(a)

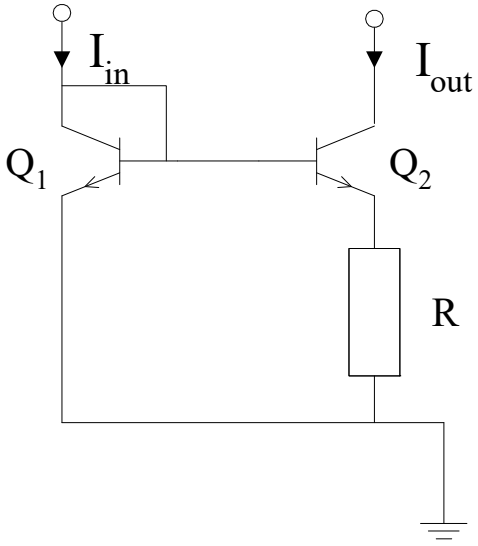


Figure 1(b)

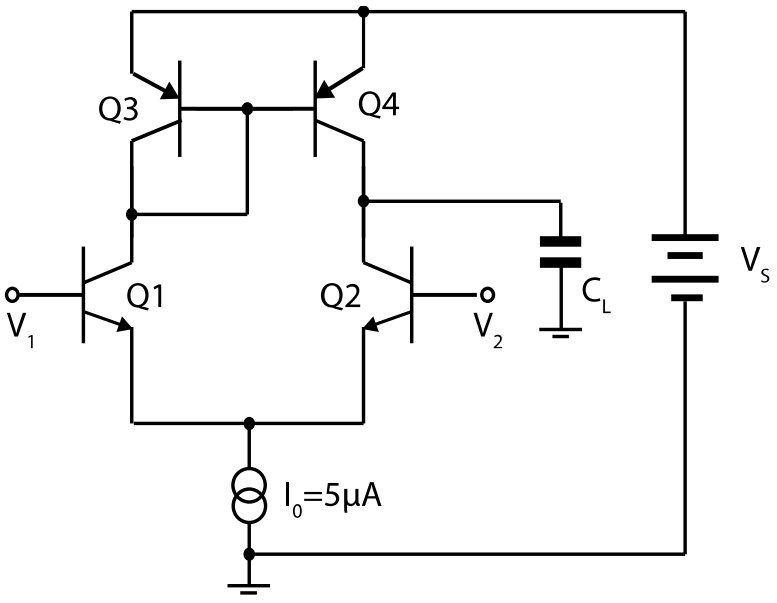


Figure 1(c)

2. (a) Figure 2(a) shows a folded cascode connection. What is the main advantage of this design over the more classical cascode connection? Show via a brief sketch how the architecture

of Figure 2(a) can be used to form the basis of a single stage fully differential folded cascode operational amplifier (op-amp) with common-mode feedback included.

[6]

(b) Figure 2(b) shows a two-stage CMOS op-amp.

(i) Estimate the low-frequency differential voltage gain and gain-bandwidth product of the amplifier. Aspect ratios of all devices are shown on the circuit. Assume all bulk effects are negligible. Device model parameters are given below.

(ii) What is the main advantage and disadvantage of a single-stage compared to a two-stage op-amp ?

[8]

(c) Explain why a resistor in series with the compensation capacitor C_c in Figure 2(b) can significantly improve the amplifier's phase margin.

[3]

(d) Draw an equivalent circuit for the PSRR of the op-amp and explain why it is low.

[3]

CMOS TRANSISTOR PARAMETERS

MODEL PARAMETERS	*Kp* ($\mu A/V^2$)	λ (V^{-1})	V_{T0} (V)
PMOS	20	0.03	-0.8
NMOS	30	0.02	1.0

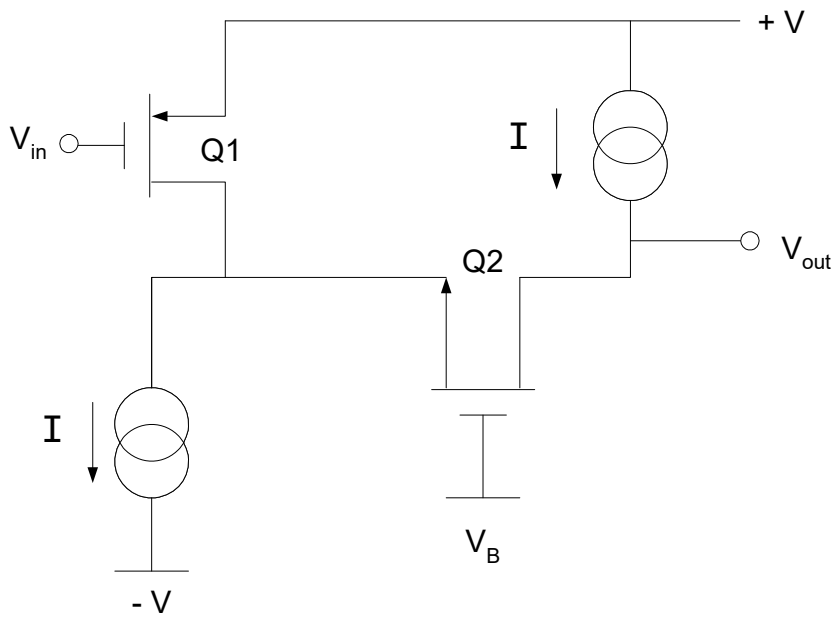


Figure 2(a)

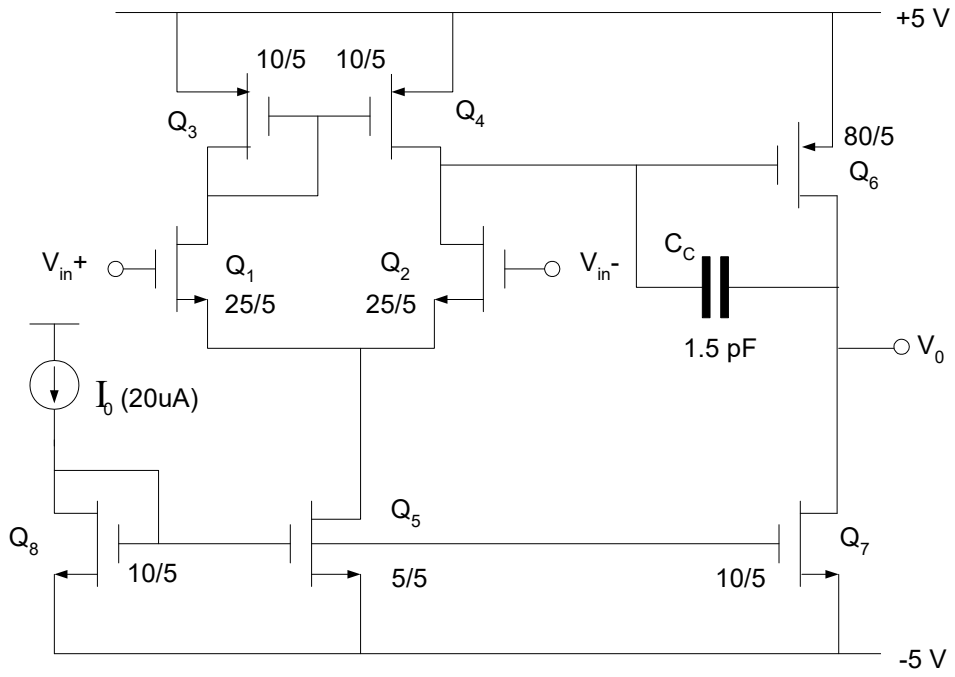


Figure 2(b)

3. (a) Sketch circuits for three integrators – A fully differential continuous-time integrator, a switched-capacitor integrator and a switched-current integrator.

[4]

(b) Explain the advantages and disadvantages of each of the integrators.

[6]

(c) (i) Derive the integrator transfer function of each of the integrators.

(ii) For both the switched-capacitor and switched-current integrators, what limits the maximum operating frequency?

[10]

4. Figure 4 shows a differential continuous-time integrator which is based upon a single-stage fully differential op-amp.

(a). Sketch the circuit diagram of a single-stage fully differential op-amp including common mode feedback.

[6]

(b). Explain the technique used for efficient layout when designing transistor-based amplifiers.

[4]

(c). (i) Transistors Q1 and Q2 of Figure 4 could be considered to form a linear differential resistor. Derive the equation for linear differential resistance. How does the body effect affect the linearity of each of these transistors?

[5]

(ii) Assuming the time constant of the circuit is τ=10ms, calculate the value of feedback capacitor, C, assuming input transistors Q1 and Q2 are operating in the triode region and have W/L=5, K=20uA/V2, Vg=1V, Vt=0.5V and node x is at a common mode of 0 V. You may neglect any body effects.

[5]

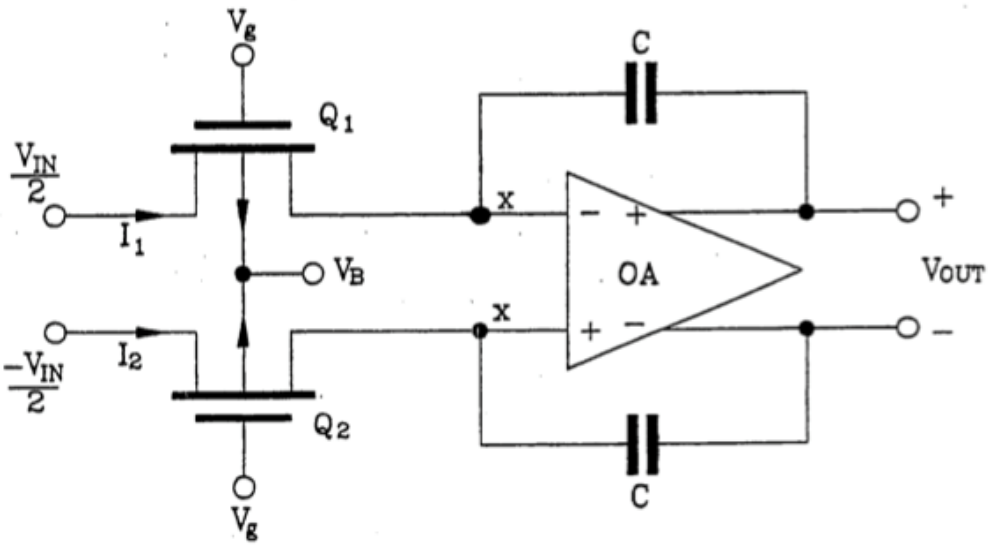


Figure 4.

5. (a) Give two advantages of current-mode analogue signal processing compared to traditional voltage-mode processing.

[2]

(b) With the aid of a suitable macromodel, explain the theoretical concept of current feedback and how it results in constant bandwidth amplification. Using a current-feedback op-amp, design a closed-loop non-inverting gain stage with a bandwidth of 10 MHz for a fixed voltage gain of 100. Assume an internal compensation capacitance of 4pF and that the open-loop transresistance gain of the amplifier is very much larger than the amplifier feedback resistor.

[6]

(c) In order to improve the voltage gain of single stage amplifiers the cascading technique was introduced.

Please sketch suitable circuits for a single cascode amplifier, a regulated cascode amplifier and a high swing cascode amplifier and derive the output resistance of each.

[6]

(d) Sketch a typical architecture of a current-mode algorithmic analogue to digital converter and explain its principles of operation.

[6]

6. Figures 6(a) and 6(b) show the basic design of two integrated circuit precision integrators.

   (a) (i) Derive an expression for the time constant of each integrator. Assume that the sampled-data integrator of Figure 6(b) is driven by non-overlapping clocks and that the switches are ideal.

   (ii) Give advantages and disadvantages of continous-time versus sampled data integrators.

   [8]

   (b) With the addition of a single switched capacitor to Figure 6(b) show how the integrator can be made lossy?

   [4]

   (c) Show how the integrator of Figure 6(b) can be used to mimic both the inductor and capacitor of an LCR passive ladder filter.

   [8]

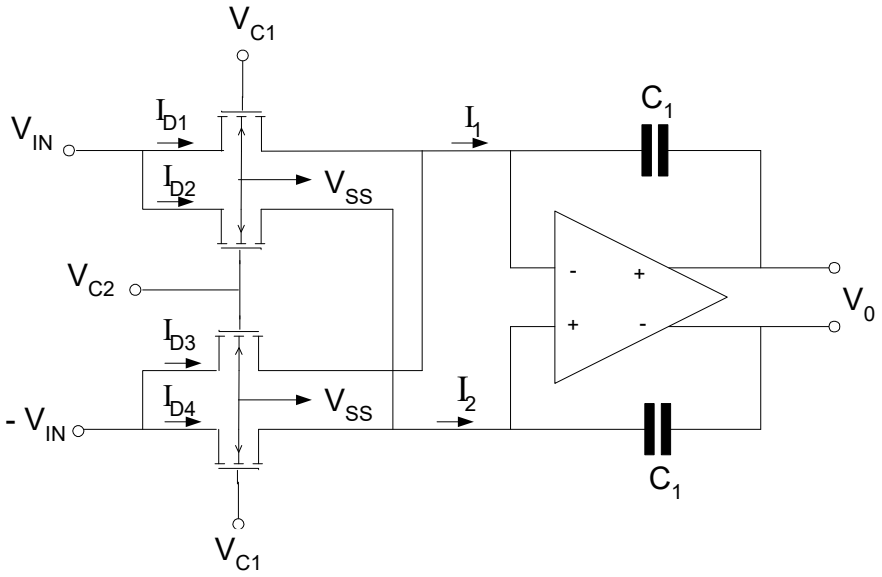


Figure 6(a)

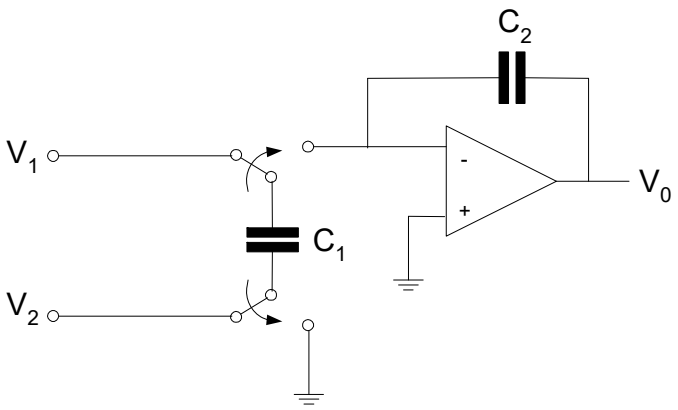


Figure 6(b)